$V_{id} = V_+ - V_-$

EECS311 Mid-1

## Diodes

### Constant model

$\begin{cases} V_D \geq V_{on},\ I_D > 0 \text{ (short)} \\ V_D < V_{on},\ I_D = 0 \text{ (open)} \end{cases}$ ($V_{on} = 0.7V$)

(Ideal model if $V_{on} = 0$)

### Exponential model

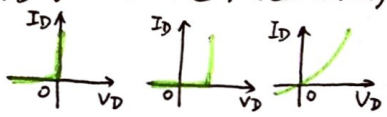
$I_D = I_S(e^{qV_D/nkT} - 1) = I_S e^{V/V_T}$

$q = 1.6\times10^{-19}C$, $k = 1.38\times10^{-23} J/K$

if $n=0$. $V_T = 0.0862T$. (mV)

$= \begin{cases} 25.8mV\ (25^\circ C) \\ 25.3mV\ (20^\circ C) \end{cases}$

$\begin{cases} I_S \text{ doubles } 5^\circ C\uparrow\ (I_R\ 10^\circ C) \\ V_D \downarrow 2mV.\ 1^\circ C\uparrow\ (\text{const } I_S) \end{cases}$




### Diode Circuits

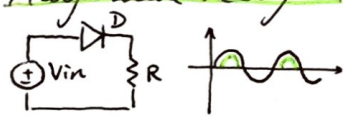
① Assume Region ② Solve

③ Check!!! (也可用图象)

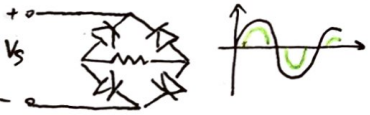
* Constant model (Ideal)

$\begin{cases} \text{不通则看作断路} \\ \text{通则看作反向电压源} \\ \quad (\text{短路}) \end{cases}$

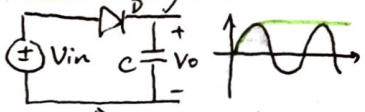
### Half-wave rectifier

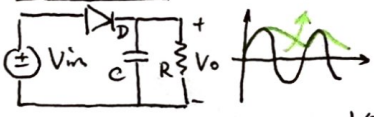



If $V_{in} = k\sin t$, $\langle V_{out}\rangle = \frac{k}{\pi}$.




### Peak Rectifier



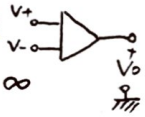

$\Delta t \ll T$, $I_L = V_P/R$, $V_r = \frac{V_P}{fCR}$

$i_{Dav} = I_L(1+\pi\sqrt{2V_P/V_r})$

$i_{Dmax} = I_L(1+2\pi\sqrt{2V_P/V_r})$

$V_o = V_P - \frac{1}{2}V_r$

### Ideal Op-amp




① $A_o = V_o/V_{id} \to \infty$

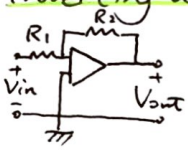
$\Rightarrow V_+ = V_-$

② $R_{in} = \infty \Rightarrow i_+ = i_- = 0$

(output current $\neq 0$)

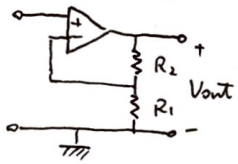
③ $R_{out} = 0 \Rightarrow$ no voltage ↓

### Inverting amplifier




$\frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1}$

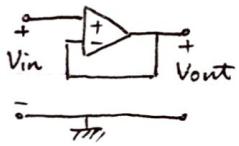
### Non-inverting amplifier




$\frac{V_{out}}{V_{in}} = \frac{A_o}{1+A_o f} \xlongequal{A_o\to\infty} \frac{1}{f} = \frac{R_1+R_2}{R_1}$

feedback factor. $:= \frac{V_-}{V_o}$

### Voltage follower

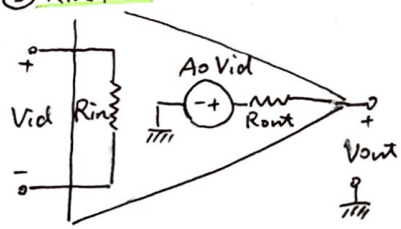



$\frac{V_{out}}{V_{in}} = \frac{A_o}{1+A_o} \xlongequal{A\to\infty} 1$

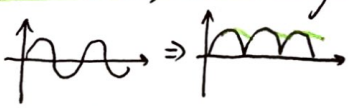
### Non-ideal Op-amp

① $A_o \neq \infty$ ② $R_{out} \neq 0$

③ $R_{in} \neq \infty$




### (Full-wave) Peak Rectifier



$V_r = \frac{V_P}{2fCR}$.

$i_{Dav} = I_L(1+\pi\sqrt{V_P/2V_r})$

$i_{Dmax} = I_L(1+2\pi\sqrt{V_P/2V_r})$

increased C.

increased R.

decreased T.

A has no impact

④ Common mode rejection ratio (CMRR)

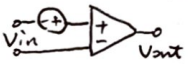
$\begin{cases} V_{id} = V_+ - V_- \\ V_{ic} = (V_+ + V_-)/2 \end{cases}$

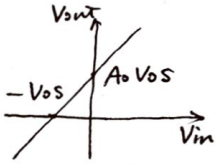
$V_{out} = A_o V_{id} + A_{cm} V_{ic}$

$CMRR = A_o/A_{cm}$

e.g. $\frac{V_{out}}{V_{in}} = \frac{A_o(1+\frac{1}{2}CMRR)}{1+A_o(1-\frac{1}{2}CMRR)}$

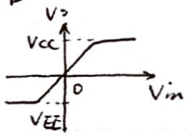
⑤ Input offset voltage



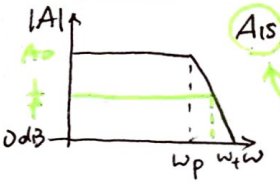




⑥ Input bias current

⑦ Clipping




⑧ Bandwidth




$A(s) = \frac{A_o}{1+s/\omega_p}$

open-loop gain $:= \frac{V_o}{V_{id}}$

If $\omega_t \gg \omega_p$, $\omega_t = A_o\omega_p$

$\Rightarrow f_t = A_o f_p$

unity gain bandwidth = gain-bandwidth product.

⑨ Slew Rate $= \left.\frac{\partial V_{out}}{\partial t}\right|_{max}$

### Sallen & Key

$H(s) = \frac{a_2s^2+a_1s+a_0}{b_2s^2+b_1s+b_0}$

$= h\frac{N(s)}{1+d(\frac{s}{\omega_0})+(\frac{s}{\omega_0})^2}$

$d = \frac{b_1}{\sqrt{b_0b_2}}$, $\omega_0 = \sqrt{\frac{b_0}{b_2}}$.

$s_1, s_2 = \omega_0\frac{-d\pm\sqrt{d^2-4}}{2}$

$T_1 = R_1C_1\omega_0$, $T_2 = R_2C_2\omega_0$

$\rho = R_1/R_2$, $\gamma = C_2/C_1$

### Bode Plot

① Find $H(s)$ for $s\to\infty, 0$

② Find poles & zeros

| | $\lvert H(j\omega)\rvert$ | $\angle H(j\omega)$ |
|---|---|---|
| pole | +20dB/dec | +45°/dec |
| zero | −20dB/dec | −45°/dec |

turning (±idea)

* quadratic case

$\begin{cases} d=0 & \text{undamped} \\ 0<d<2 & \text{underdamped} \\ d=2 & \text{critically damped} \\ d>2 & \text{overdamped.} \end{cases}$

$Q = 1/d$.

### Laplace Transform

$F(s) = \mathcal{L}[f(t)] = \int_{t=0^-}^{\infty} f(t)e^{-st}dt$

$f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j}\int_{\sigma-j\omega}^{\sigma+j\omega} F(s)e^{st}ds$

### Initial Value Theorem

$f(0^+) = \lim_{s\to\infty} sF(s)$

### Final Value Theorem

$f(\infty) = \lim_{s\to 0} sF(s)$

$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s}$

$v(t) = v_0\cos(\omega t+\varphi)$

$= Re\{V_0e^{j(\omega t+\varphi)}\}$

$\Leftrightarrow \hat{V} = V_0\angle\phi$

$V_0 = |V(\omega)| = v_0$

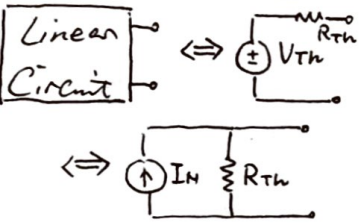
$\phi = \angle V(\omega)$

Euler's Formula:

$\boxed{e^{j\theta} = \cos\theta + j\sin\theta}$

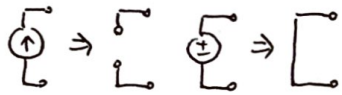
| | | |
|---|---|---|
| R | $V=IR$ | $Z=R$ |
| L | $V=L\frac{di}{dt}$ | $Z=sL$ |
| C | $i=C\frac{dv}{dt}$ | $Z=\frac{1}{sC}$ |

## Thévenon & Norton




$I_N = V_{Th} / R_{Th}$

* $V_{Th}$: open-loop voltage
* $I_N$: short-loop current
* $R_{Th}$: ① zero out independent sources;



leave dependent ~.

② $R_{Th} = V_T / I_T$

## Step Response

For one-stage low-pass filter $V(t) = V_0(1 - e^{-\frac{t}{\tau}})$

rise time: $t_r = \tau \cdot \ln 9$

(time needed from 10% ~ 90%)

## NMOS

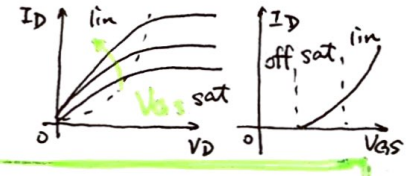
① Cutoff $I_D = 0$. $V_{GS} < V_{Th}$

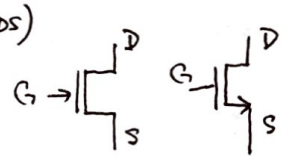
② linear. $V_{GS} > V_{Th}$. $V_{DS} \le V_{GS} - V_{Th} = V_{ov}$

$I_D = K_n [V_{ov} V_{DS} - \frac{1}{2} V_{DS}^2]$

③ saturation $V_{GS} > V_{Th}$. $V_{DS} > V_{ov}$

$I_D = \frac{1}{2} K_n (V_{GS} - V_{Th})^2 (1 + \lambda V_{DS})$




## Laplace Transform Properties

$\delta(t) \longleftrightarrow 1$

$\delta(t-T) \longleftrightarrow e^{Ts}$

$1/u(t) \longleftrightarrow \frac{1}{s}$

$u(t-T) \longleftrightarrow \frac{e^{-Ts}}{s}$

$e^{-at}u(t) \longleftrightarrow \frac{1}{s+a}$

$e^{-a(t-T)}u(t-T) \longleftrightarrow \frac{e^{-Ts}}{s+a}$

$t\,u(t) \longleftrightarrow \frac{1}{s^2}$

$(t-T)u(t-T) \longleftrightarrow \frac{e^{-Ts}}{s^2}$

$t^2 u(t) \longleftrightarrow \frac{2}{s^3}$

$te^{-at}u(t) \longleftrightarrow \frac{1}{(s+a)^2}$

$t^2 e^{-at}u(t) \longleftrightarrow \frac{2}{(s+a)^3}$

$t^{n-1}e^{-at}u(t) \longleftrightarrow \frac{(n-1)!}{(s+a)^n}$

$\sin\omega t\, u(t) \longleftrightarrow \frac{\omega}{s^2+\omega^2}$

$\sin(\omega t+\theta)u(t) \longleftrightarrow \frac{s\sin\theta + \omega\cos\theta}{s^2+\omega^2}$

$\cos\omega t\, u(t) \longleftrightarrow \frac{s}{s^2+\omega^2}$

$e^{-at}\sin\omega t\, u(t) \longleftrightarrow \frac{\omega}{(s+a)^2+\omega^2}$

$e^{-at}\cos\omega t\, u(t) \longleftrightarrow \frac{s+a}{(s+a)^2+\omega^2}$

$2e^{-at}\cos(bt-\theta)u(t) \longleftrightarrow \frac{e^{j\theta}}{s+a+jb} + \frac{e^{-j\theta}}{s+a-jb}$

$\frac{2t^{n-1}}{(n-1)!}e^{-at}\cos(bt-\theta)u(t) \longleftrightarrow \frac{e^{j\theta}}{(s+a+jb)^n} + \frac{e^{-j\theta}}{(s+a-jb)^n}$

$\cos(\omega t+\theta)u(t) \longleftrightarrow \frac{s\cos\theta - \omega\sin\theta}{s^2+\omega^2}$

* $\frac{k}{(\beta-\alpha)}(e^{-\alpha t} - e^{-\beta t})u(t) \longleftrightarrow \frac{k}{(s+\alpha)(s+\beta)}$

## Laplace Transform

$kf(t) \longleftrightarrow kF(s)$

$k_1 f_1(t) + k_2 f_2(t) \longleftrightarrow k_1 F_1(s) + k_2 F_2(s)$

$f(at), a>0 \longleftrightarrow \frac{1}{a}F(\frac{s}{a})$

$f(t-T)u(t-T) \longleftrightarrow e^{-Ts}F(s),\ T \ge 0$

$e^{-at}f(t) \longleftrightarrow F(s+a)$

$f' = \frac{df}{dt} \longleftrightarrow sF(s) - f(0^-)$

$f'' = \frac{d^2f}{dt^2} \longleftrightarrow s^2F(s) - sf(0^-) - f'(0^-)$

$\int_{0^-}^{t} f(t')dt' \longleftrightarrow \frac{1}{s}F(s)$

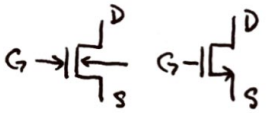
$tf(t) \longleftrightarrow -\frac{d}{ds}F(s)$

$\frac{f(t)}{t} \longleftrightarrow \int_s^\infty F(s')ds'$

# EECS311 Mid-2

## MOSFET

### NMOS



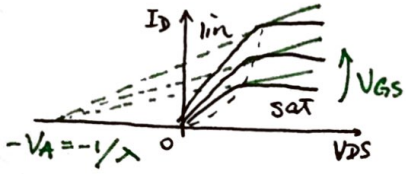

① Cutoff $V_{GS} < V_{tn}$. $I_D = 0$

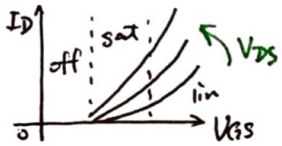
② linear $V_{GS} > V_{tn}$, $V_{DS} \le V_{GS} - V_{tn} \equiv V_{ov}$

$I_D = K_n(V_{ov}V_{DS} - \frac{1}{2}V_{DS}^2)$ * $K_n = K'W/L$, $K' = \mu_n C_{ox}$

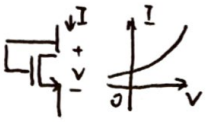
③ saturation $V_{GS} > V_{tn}$, $V_{DS} > V_{ov}$

$I_D = \frac{1}{2}K_n V_{ov}^2(1+\lambda V_{DS})$ * $\lambda = \Delta L/L$ (C.L.M)



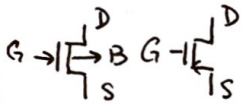




diode-connected transistor




sat $\Rightarrow I_D = \frac{1}{2}KV_{ov}^2 = \frac{1}{2}K(V-V_{tn})^2$

### PMOS




① Cutoff $V_{GS} > V_{tp}$ $I_D = 0$

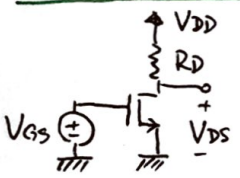
② linear $V_{GS} < V_{tp}$, $V_{DS} \ge V_{GS} - V_{tp} \equiv V_{ov}$

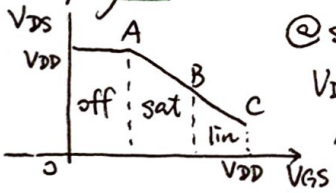
$I_D = -k_p(V_{ov}V_{DS} - \frac{1}{2}V_{DS}^2)$

③ saturation $V_{GS} < V_{tp}$, $V_{DS} < V_{ov}$

$I_D = -\frac{1}{2}k_p V_{ov}^2(1+\lambda V_{DS})$

### Common Source Amplifier



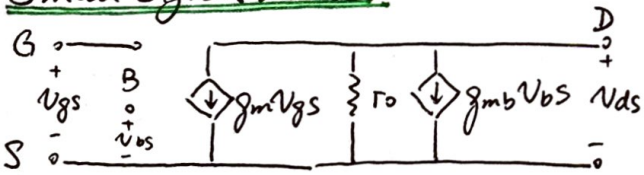




@sat

$V_{DS} = V_{DD} - \frac{1}{2}R_D K_n V_{ov}^2$

$A_v = -g_m R_D$

### Small Signal Model

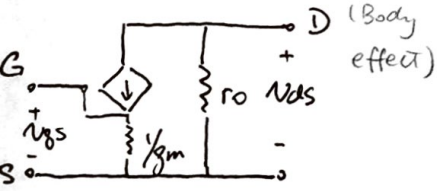



Hybrid-π model

$g_{m,sat} = \frac{\partial i_{D,sat}}{\partial v_{GS}}\Big|_{v_{GS}=V_{GS}} = K_n(V_{GS} - V_{tn}) = \sqrt{2K_n I_{D,sat}} = -\frac{2R_D I_D}{V_{ov}}$

$r_o = \left(\frac{\partial i_D}{\partial v_{DS}}\right)^{-1} = V_A/I_D$ (CLM)

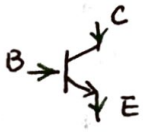
$g_{mb} = \left(\frac{\partial i_D}{\partial v_{BS}}\right) = \frac{\gamma}{2\sqrt{2\phi_F + V_{SB}}}$




T-model

## BJT

### NPN



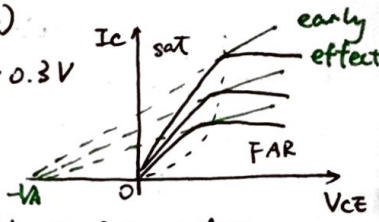

① Cutoff EB.CB reversed

② FAR EB forward, CB reversed

$\alpha = i_C/i_E$, $\beta = i_C/i_B$, $i_C = I_S e^{V_{BE}/V_T}$, $V_T = 25.8mV$

$\alpha = \beta/(\beta+1) \Leftrightarrow \beta = \alpha/(1-\alpha)$

$V_{BE} = 0.7V$, $V_{BC} < 0.4V$, $V_{CE} > 0.3V$



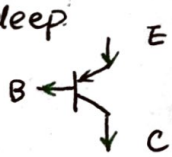

③ sat EB.CB forward

$\beta_{forced} \le \beta$. $i_C \downarrow$. $i_b \uparrow$

$V_{BE} = 0.7V$ | $V_{BC} = 0.4V$, $V_{CE} = 0.3V$ edge

| $V_{BC} = 0.5V$, $V_{CE} = 0.2V$ deep

### PNP

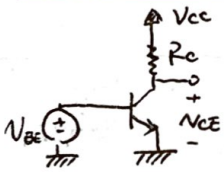


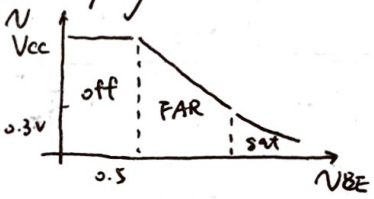

Simply swap subscripts of NPN.

Early Effect: $\beta_F = \beta_{F0}(1 + V_{CE}/V_A)$

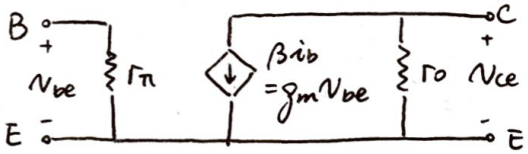
$i_C = I_S e^{V_{BE}/V_T}(1 + V_{CE}/V_A)$

### Common Emitter Amplifier




### Small Signal Model



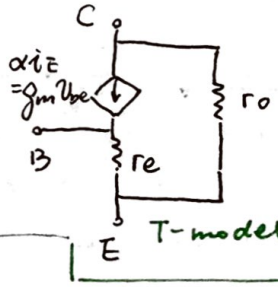

Hybrid-π model

$g_m = I_C/V_T$

$r_o = (V_A + V_{CE})/I_C = V_A/I_C$

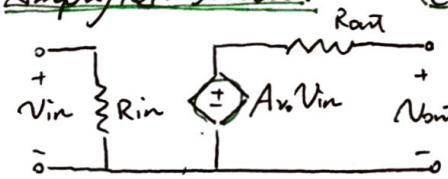
$r_\pi = \beta/g_m = V_T/I_B = \beta V_T/I_C$

$r_e = r_\pi/(\beta+1) = V_T/I_E = \alpha/g_m$




T-model

### Amplifier Model



For common source amp

$R_{in} = \infty$, $R_{out} = r_o // R_D$

$A_{vo} = -g_m(R_D // r_o)$

For common emitter amp

$R_{in} = r_\pi$, $R_{out} = r_o // R_C$

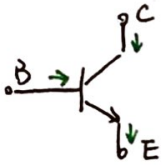
$A_{vo} = -g_m(R_C // r_o)$

### Overall voltage gain

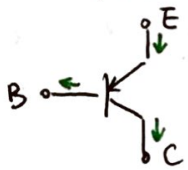
$G_v = R_{in}/(R_{in} + R_{sig}) \cdot A_{vo} \cdot R_L/(R_L + R_{out})$

# Large Signal Mode of BJT

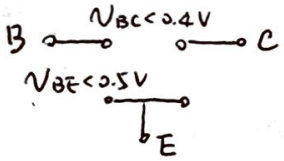
NPN

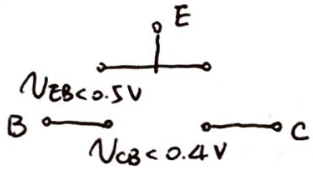



PNP

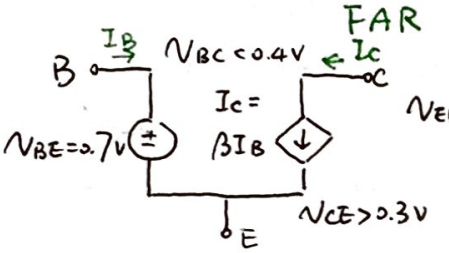


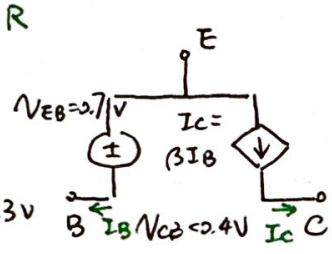

cut off



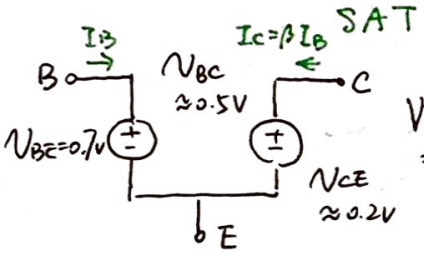



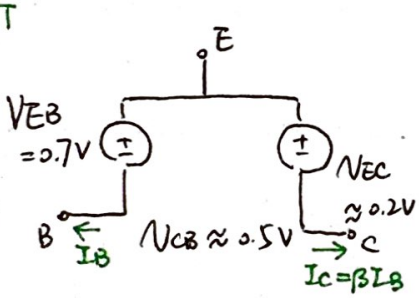

FAR



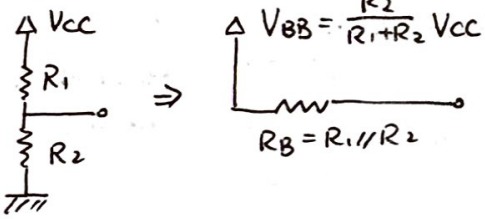




SAT

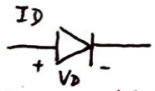


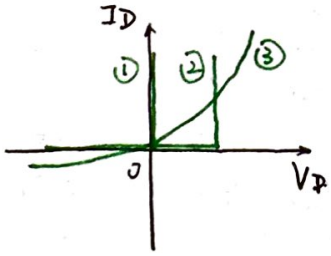




## Symplified model




## Diode




① ideal $\begin{cases} I_D>0,\ V_D\geq 0 \\ I_D=0,\ V_D<0 \end{cases}$

② constant voltage $\begin{cases} I_D>0,\ V_D\geq V_{ON} \\ I_D=0,\ V_D<V_{ON} \end{cases}$

③ exponential $I_D=I_S(e^{qV_D/nkT}-1)$

$=I_S e^{V/V_T}$ (if $i\gg I_S$)

$V_T=25.8mV$ (@ room tempreture)

# EECS311 Final

## MOSFET

**NMOS**

① off $V_{GS} < V_{tn}$ $I_D = 0$

② lin $V_{GS} > V_{tn}$

$V_{DS} \le V_{GS} - V_{tn} = V_{ov}$

$I_D = k_n(V_{ov}V_{DS} - \frac{1}{2}V_{DS}^2)$

③ sat $V_{GS} > V_{tn}$, $V_{DS} > V_{ov}$

$I_D = \frac{1}{2}k_n V_{ov}^2(1+\lambda V_{DS})$

* $k = k'W/L$ $k' = \mu C_{ox}$

**PMOS**

$V_{GS} > V_{tp}$ $I_D = 0$

$V_{GS} < V_{tp}$

$V_{DS} \ge V_{GS} - V_{tp} = V_{ov}$

$I_D = -k_p(V_{ov}V_{DS} - \frac{1}{2}V_{DS}^2)$

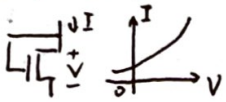
$V_{GS} < V_{tp}$, $V_{DS} < V_{ov}$

$I_D = -\frac{1}{2}k_p V_{ov}^2(1+\lambda V_{DS})$

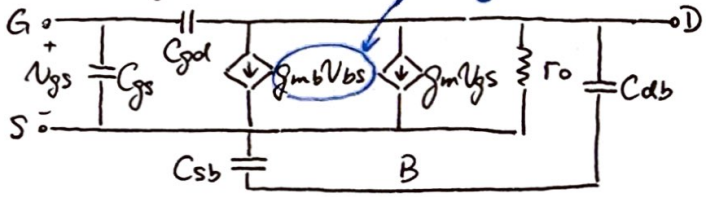
* $\lambda = \Delta L/L$ (C.L.M)

diode-connected transistor

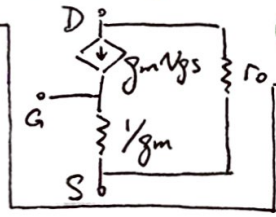
sat $\Rightarrow I_D = \frac{1}{2}k V_{ov}^2$



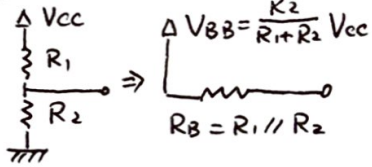
### Small Signal Model

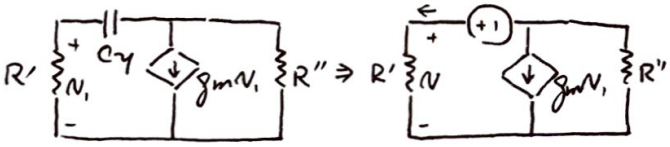



$g_m = k_n V_{ov} = \sqrt{2k_n I_D} = \frac{-2R_D I_D}{V_{ov}}$, $r_o = V_A/I_D$ (C.L.M)




### Generic Models

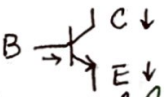



$R_{out} = R' + R''(1+g_m R')$

## BJT

**NPN**




① off EB, CB reversed $V_{BC} < 0.4V$, $V_{BE} < 0.5V$

② for EB reversed, CB forward $V_{BE} = 0.7V$ $V_{CE} > 0.3$

$\alpha = i_C/i_E$, $\beta = i_C/i_B$, $i_C = I_S e^{V_{BE}/V_T}$ $V_T = 25.8mV$ $V_{BC} < 0.4V$

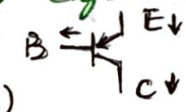
$\alpha = \beta/(\beta+1) \Leftrightarrow \beta = \alpha/(1-\alpha)$

③ sat EB, CB forward $\beta_{forced} \le \beta$. $i_C \downarrow$ $i_b \uparrow$

$V_{BE} = 0.7V$ $V_{BC} = 0.5V$ $V_{CE} = 0.2V$ deep in sat

$V_{BC} = 0.4V$ $V_{CE} = 0.3V$ edge sat/for

**PNP** swap subscripts of NPN



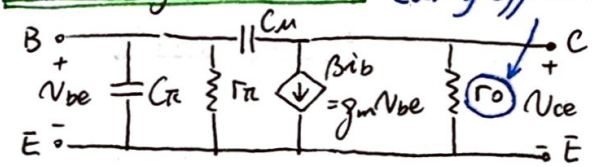

### Early Effect

$\beta_F = \beta_{F0}(1 + V_{CE}/V_A)$

$i_C = I_S e^{V_{BE}/V_T}(1 + V_{CE}/V_A)$

### Small Signal Model

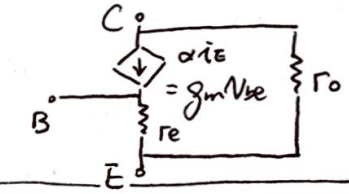
early effect (B.W.M)




$g_m = I_C/V_T$, $r_o = V_A/I_C$

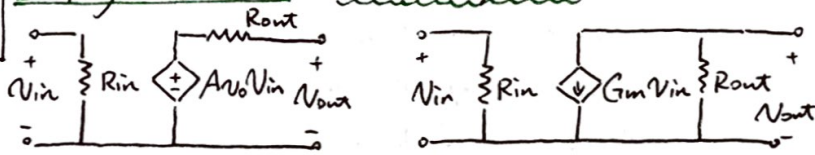
$r_\pi = \beta/g_m = V_T/I_B = \beta V_T/I_C$

$r_e = r_\pi/(\beta+1) = V_T/I_E = \alpha/g_m$




### Amplifier Model

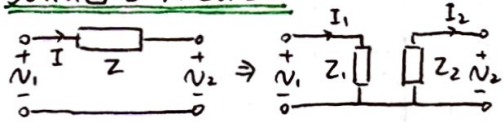
$G_m = -A_{vo}/R_{out}$




$A_v = A_{vo1}A_{vo2}\frac{R_{in1}}{R_{in1}+R_{sig}} \cdot \frac{R_2}{R_2+R_{out1}} \cdot \frac{R_L}{R_L+R_{out2}}$

$A_v = G_{m1}G_{m2}V_{in}(R_{out1} // R_{in2})(R_{out2} // R_L)$

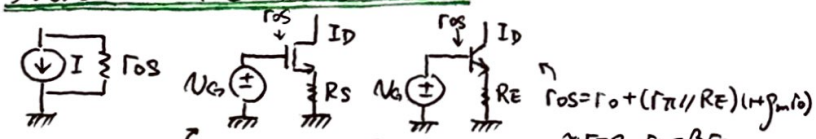
### Miller's Theorem




$K = V_2/V_1$

$Z_1 = \frac{Z}{1-K}$, $Z_2 = \frac{Z}{1-1/K}$

### Non-ideal Current Source

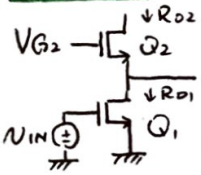



$r_{os} = r_o + R_S + r_o R_S(g_m + g_{mb})$

$r_{os} = r_o + (r_\pi // R_E)(1+g_m r_o)$

$\approx r_\pi g_m r_o = \beta r_o$

### Cascode

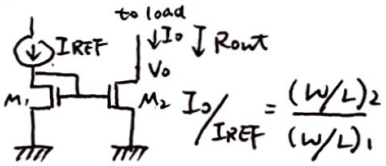



$R_{o1} = r_{o1}$

$R_{o2} = r_{o1} + r_{o2} + r_{o1}r_{o2}g_{m2}$

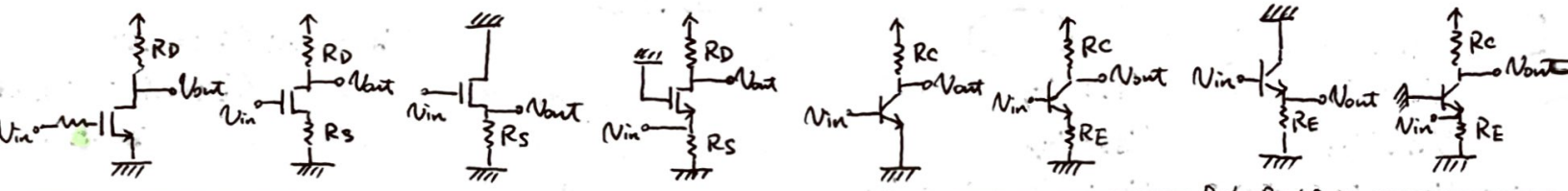
$= r_{o1} + A_{o2}$

### Current mirror




$I_o/I_{REF} = \frac{(W/L)_2}{(W/L)_1}$

$R_{out} = V_{A2}/I_o$




$R_E' = R_E // R_L // 1$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $R_{in}$ | $\infty$ l | $\infty$ l | $\infty$ l | $R_S // (\frac{1}{g_m+g_{mb}})$ s | $r_\pi$ m | $\approx r_\pi(1+g_m R_E)$ l | $(1+\beta)(r_e+R_E')$ l | $R_E // r_e$ s |
| $R_{out}$ | $R_D // r_o$ m | $R_D$ m | $R_S/(1+R_S(g_m+g_{mb}))$ s | $R_D$ m | $R_C // r_o$ m | $R_C // (r_o(1+g_m R_E))$ m | $\approx r_e // R_E'$ s | $R_C // r_o$ m |
| $A_{vo}$ | $-g_m R_D$ m | $-\frac{g_m R_D}{1+(g_m+g_{mb})R_S}$ m | $\frac{g_m R_S}{1+R_S(g_m+g_{mb})} \le 1$ | $(g_m+g_{mb})R_D$ l | $-g_m R_C$ l | $-\frac{g_m R_C}{1+g_m R_E}$ m | $\approx \frac{g_m R_E'}{1+g_m R_E'} \le 1$ | $\approx g_m R_C$ l |
| $\omega_H$ | $\approx \frac{1}{C_{gd} R_G \lvert A_{vo}\rvert}$ m | $\approx \frac{1}{C_{gd} R_G \lvert A_{vo}\rvert}$ l | large | large | $\frac{1}{C_\mu (R_{sig} // R_{in}) \lvert A_{vo}\rvert}$ s | $\frac{1}{C_\mu (R_{sig} // R_{in}) \lvert A_{vo}\rvert}$ m | $\approx \frac{1}{C_\pi [R_E' // (1/g_m)]}$ l | $\frac{1}{C_\mu R_C}$ m |

## Steps for solving BJT/MOSFET amp circuit

① DC analysis

Assume region of operation; solve; check

② ac analysis (short DC voltage src, open DC current src)

mid band $\begin{cases} A_{vo} = v_{out}/v_{in} \mid_{R_L=\infty} \\ R_{in} = v_{in}/i_{in} \\ R_{out} = v_t/i_t \text{ (input shorted)} \end{cases}$

SCTC i) open high freq caps.

short other low freq. caps.

ii) Short input. $R_{Bi} = v_t/i_t$

iii) $\omega_L = \frac{1}{R_{B1}C_{B1}} + \frac{1}{R_{B2}C_{B2}} + \dots + \frac{1}{R_{Bi}C_{Bi}}$

OCTC i) short low freq caps.

open other high freq caps

ii) short input $R_{oi} = v_t/i_t$

iii) $\omega_H = \frac{1}{R_{o1}C_{o1} + R_{o2}C_{o2} + \dots + R_{oi}C_{oi}}$